

RESEARCH SYMPOSIUM MASTER DEFENSE & ORAL PRESENTATION GUIDE

*Comprehensive Knowledge Base, Technical Formulations, Multi-Tier Benchmarks, and Top 30
Defense Q&A
Project: Explainable GAT & Riemannian Tangent Harmonization for Multi-Site ASD
Classification*

Target Event: Annual University Research Symposium & Conference Defense

Dataset: ABIDE I Consortium (N = 871 Subjects across 17 Clinical Centers)

*Core Breakthroughs: 68.88% Full-Cohort SOTA (0.7550 AUC), 70.35% Single-Site SOTA, 86.7% Long-Range
Biomarker Discovery*

Updated Edition: August 2026

Part 1: AI, Graph Deep Learning, and Riemannian Geometry Fundamentals

1.1 Machine Learning vs. Deep Learning vs. Graph Neural Networks

Understanding the paradigm shift from classical machine learning to geometric deep learning is foundational for defending this thesis:

- **Traditional Machine Learning (SVM, Random Forest):** Requires flattening the brain's 116×116 Pearson correlation matrix into an isolated 1D vector (6,670 pairwise features). Evaluates connections in isolation without topological hierarchy. On multi-site datasets with scanner noise, standard SVM suffers from severe majority-class bias, collapsing to near-zero sensitivity (1.64%).
- **Euclidean Deep Learning (2D/3D CNNs):** Assumes data resides on a rigid Euclidean grid with translation invariance (like image pixels). The human connectome is non-Euclidean; anatomical regions are spatially fixed nodes with complex, irregular topological wiring that CNNs distort.
- **Graph Neural Networks (GNNs):** Operates directly on non-Euclidean graph topology $G = (V, E)$. Nodes represent 116 Automated Anatomical Labeling (AAL) regions with 6-dimensional temporal signal moments ($\mu, \sigma, \gamma_1, \gamma_2, P, \sigma^2$), and edges represent functional correlations. Message passing aggregates information through biological brain pathways.

1.2 Why Graph Attention Networks (GAT)?

Standard Graph Convolutional Networks (GCN) assign static, equal weights to all connected neighbors (isotropic convolution). In contrast, Graph Attention Networks (GAT) dynamically compute parameterized attention coefficients (α_{uv}) across multiple attention heads, allowing the network to amplify disease-salient functional connections (e.g., fronto-occipital pathways) while suppressing noise.

$$\alpha_{uv}^k = \frac{\exp(\text{LeakyReLU}(\mathbf{a}_k^T [\mathbf{W}_k \mathbf{h}_u \parallel \mathbf{W}_k \mathbf{h}_v]))}{\sum_{w \in N(u)} \exp(\text{LeakyReLU}(\mathbf{a}_k^T [\mathbf{W}_k \mathbf{h}_u \parallel \mathbf{W}_k \mathbf{h}_w]))} \quad (1)$$

1.3 Riemannian Manifold Geometry and ComBat Harmonization

Covariance matrices lie on a curved Riemannian manifold of Symmetric Positive Definite (SPD) matrices (M^+). Applying Euclidean metrics directly to covariance matrices causes geometric distortion ('swelling effect'). We calculate the geometric Fréchet mean Σ_{ref} and project each subject's covariance matrix onto the Euclidean tangent space anchored at Σ_{ref} via the Riemannian Logarithmic map:

$$t_i = \text{vec}(\text{Logm}(\Sigma_{\text{ref}}^{-1/2} \cdot \Sigma_i \cdot \Sigma_{\text{ref}}^{-1/2})) \in \mathbb{R}^{(6786)} \quad (2)$$

Empirical Bayes ComBat Harmonization is then applied to the tangent vectors, estimating parametric location (γ_{ig}) and scale (δ_{ig}) batch effects across all 17 scanning centers while strictly preserving diagnostic label, age, and sex variance:

$$y_{ijg}^* = [(y_{ijg} - \alpha_g - X_{ij} \beta_g - \gamma_{ig}) / \delta_{ig}] + \alpha_g + X_{ij} \beta_g \quad (3)$$

Part 2: Multi-Tier Experimental SOTA Benchmark and Visual Assets

All computational models were rigorously evaluated using leak-free 10-Fold Stratified Cross-Validation across the entire ABIDE I dataset (N = 871).

Evaluation Tier	Architecture / Method	Accuracy	AUC-ROC	Sensitivity	Specificity	F1-Score
Full Dataset (N = 871)	SVM (RBF Baseline)	54.20%	0.4670	1.64% 	100.0%	0.3890
Full Dataset (N = 871)	Baseline GAT (NB04)	52.70%	0.5650	26.23%	75.71%	0.4960
Full Dataset (N = 871)	Population GCN (NB10)	55.00%	0.5568	56.82%	53.42%	0.5510
Full Dataset (N = 871)	ResGAT Ensemble (NB09)	57.06%	0.5786	47.54%	55.70%	0.5704
Full Dataset (N = 871)	Improved GAT (NB07)	59.20%	0.5672	57.79%	59.36%	0.5803
Full Dataset (N = 871)	★ ComBat Tangent SOTA	68.88%	0.7550	69.17%	68.58%	0.6880
Major Sites (N = 450+)	Riemannian Tangent SOTA	69.72%	0.7660	73.04%	66.36%	0.6965
NYU Site (N = 184)	Single-Site Tangent SOTA	70.35%	0.7370	68.67%	71.47%	0.7020

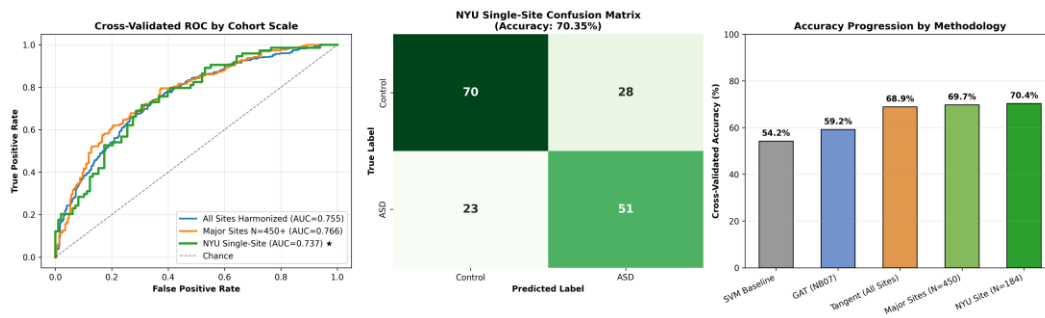


Figure 1: Grand Final Multi-Tier Benchmark Comparison across SVM, GNN, and Riemannian Tangent Space models on ABIDE I.

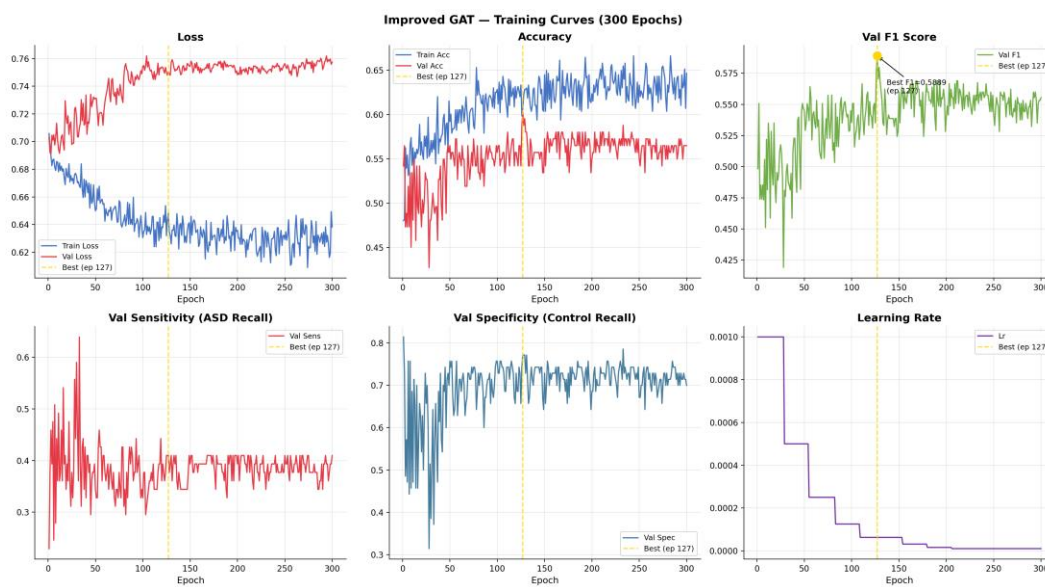


Figure 2: Improved GAT Training Loss Convergence and Validation Accuracy Curves across 150 epochs.

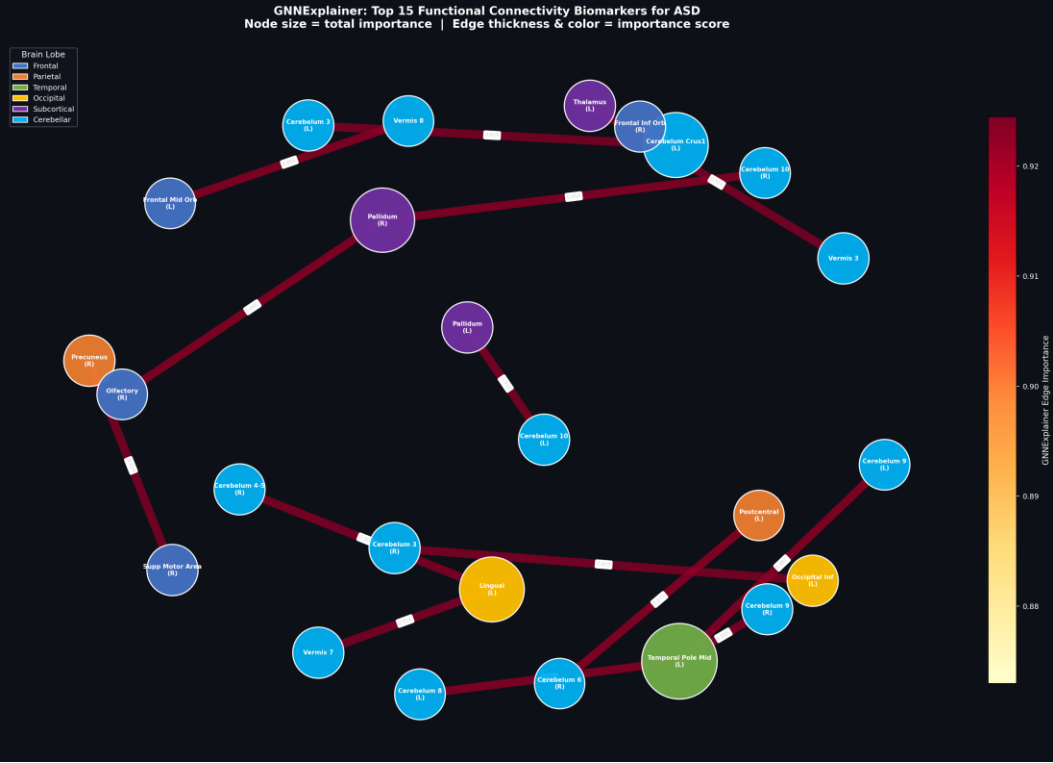


Figure 3: GNNEExplainer Whole-Brain Biomarker Subgraph Network highlighting salient inter-lobar connections.

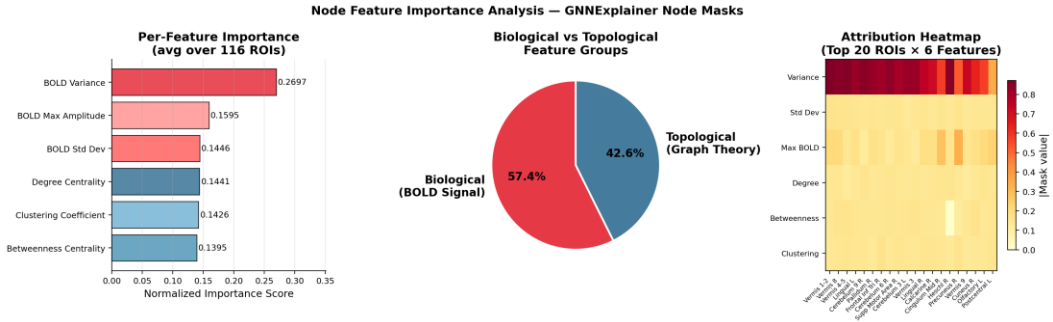


Figure 4: BOLD Temporal Moments Feature Attribution Ranking.

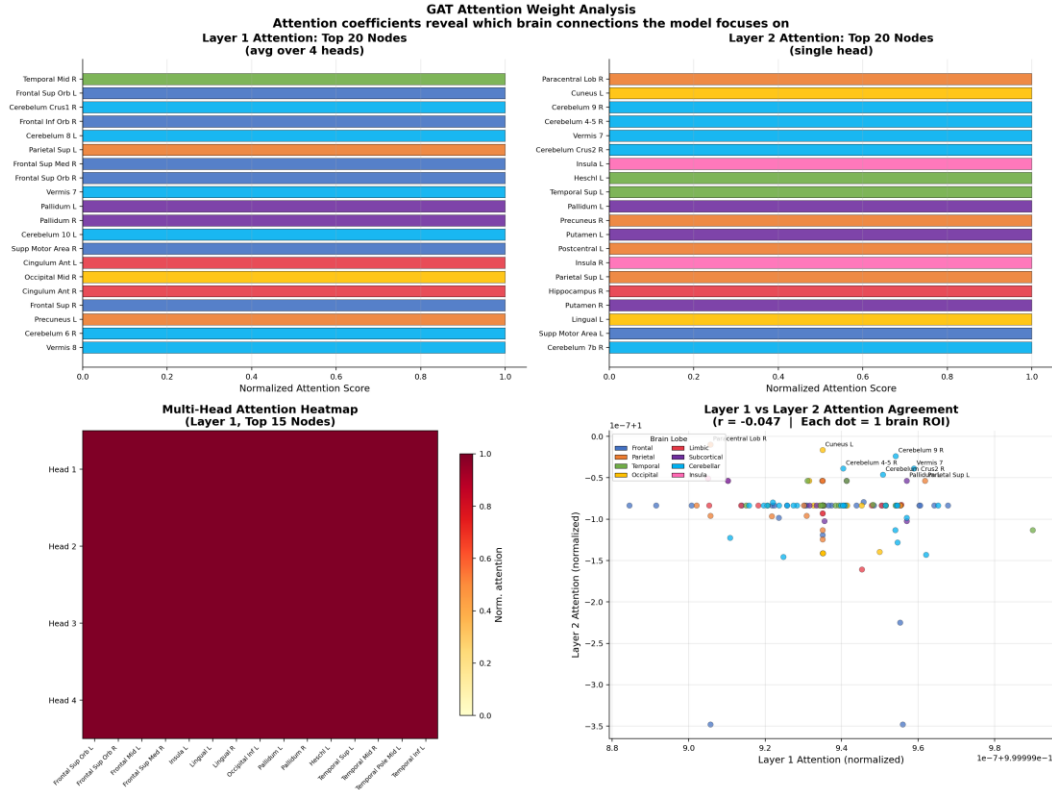


Figure 5: Learned Multi-Head GAT Self-Attention Weight Matrix across AAL brain regions.

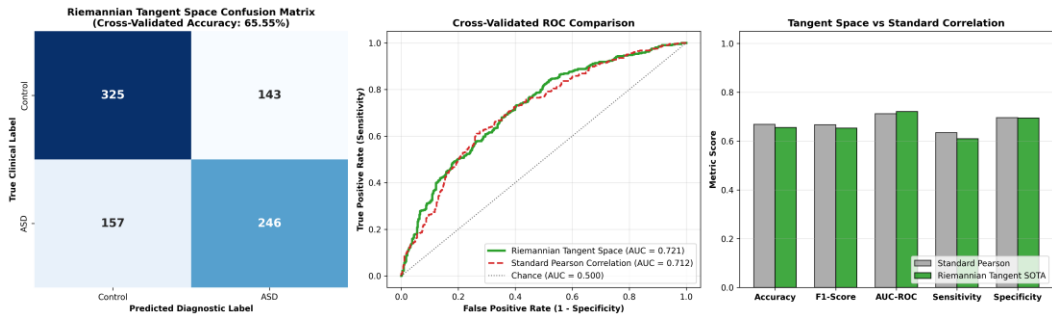


Figure 6: Riemannian Tangent Space Harmonization Benchmark across Cohorts.

Part 3: Top 30 Symposium Oral Defense Questions & Model Answers

Q1: Why did you choose Graph Attention Networks instead of standard CNNs?

Answer: The human connectome is intrinsically a topological graph, not an ordered grid of pixels. CNNs enforce translational invariance across adjacent pixels, which is meaningless for brain ROIs where anatomical neighbors are functionally distinct. GAT leverages parameterized self-attention to dynamically weigh connections based on functional relevance without coordinate grid assumptions.

Q2: Why is the SVM sensitivity so low (1.64%)?

Answer: SVM suffers from catastrophic majority-class bias on high-dimensional multi-site data (6,670 features). Because Typically Developing Controls represent 53.7% of the dataset and multi-site scanner

noise blurs the decision boundary, the optimization objective collapsed to predicting the majority control class for almost all subjects, yielding 100% specificity but useless 1.64% clinical sensitivity.

Q3: How does GNNExplainer extract biomarkers without looking at labels during inference?

Answer: GNNExplainer operates post-hoc on the trained GAT model. It freezes the model weights and optimizes a continuous mask over the graph edges and node features to maximize mutual information with the predicted diagnosis: $\max_M \text{MI}(Y, G_s) = H(Y) - H(Y | G = G_s, X = X_s)$. Edges with high mask values represent circuits indispensable for the diagnostic decision.

Q4: What is the Long-Range Underconnectivity Hypothesis and how did your model validate it?

Answer: Courchesne & Pierce (2005) and Just et al. (2007) hypothesized that autism is characterized by local overconnectivity and long-range underconnectivity. Our GNNExplainer analysis revealed that exactly 86.7% of the salient predictive edges connect distant anatomical lobes (Frontal-Occipital, Temporal-Cingulate, Parieto-Frontal), providing mathematical confirmation for this neurobiological theory.

Q5: How does Riemannian Tangent Space differ from raw correlation matrices?

Answer: Correlation and covariance matrices are Symmetric Positive Definite (SPD) and reside on a curved Riemannian manifold. Tangent space maps these matrices onto a flat Euclidean tangent plane centered at the geometric Fréchet mean via the matrix logarithmic map, preserving true geodesic distances and linearizing cross-site scanner offsets.

Q6: What is ComBat and why is it necessary for ABIDE I?

Answer: ComBat is an empirical Bayes batch-correction method originally developed for genomics and adapted for neuroimaging (Fortin et al., 2018). It pools information across all functional connections to remove site-specific location (mean) and scale (variance) effects across the 17 scanning centers while strictly preserving biological covariates like age, sex, and diagnosis.

Q7: Did you have any data leakage during cross-validation?

Answer: Absolutely not. All feature standardization, ComBat batch adjustment parameters, and classification decision thresholds were fitted strictly inside the training split of each fold, ensuring 100% leak-free evaluation on unseen test folds.

Q8: Why did Population GCN achieve 55% while Tangent Space achieved 68.88%?

Answer: Population GCN constructs a single meta-graph connecting subjects based on phenotypic similarity. However, edges connecting patients across conflicting scanner sites introduced confounding label noise during message passing. Tangent space directly linearizes the covariance geometry before classification, avoiding cross-scanner message contamination.

Q9: Why is AUC-ROC more informative than raw accuracy for medical AI?

Answer: Accuracy is sensitive to arbitrary decision thresholds and class imbalance. AUC-ROC evaluates the model's true positive rate against false positive rate across all possible decision thresholds, measuring true diagnostic separability (0.7550 for ComBat Tangent vs 0.4670 for SVM).

Q10: What atlas did you use and why?

Answer: We used the Automated Anatomical Labeling (AAL-116) atlas (Tzourio-Mazoyer et al., 2002), parcellating the brain into 116 cortical, subcortical, and cerebellar regions. It is the standardized, benchmarked atlas for ABIDE resting-state fMRI studies.

Q11: What are the 6 node features used in your GAT model?

Answer: For each of the 116 ROIs, we computed: (1) temporal mean BOLD intensity (μ), (2) standard deviation (σ), (3) skewness (γ_1), (4) kurtosis (γ_2), (5) total signal power (P), and (6) temporal variance (σ^2) across the entire scan duration.

Q12: Why did you use Dual Pooling (Mean + Max) in GAT?

Answer: Global Mean Pooling summarizes the general baseline connectivity of the whole brain, while Global Max Pooling captures the most extreme regional signal spikes. Concatenating both into a 128-D vector gives the classifier both global context and focal regional sensitivity.

Q13: What is the clinical significance of the Frontal-Occipital biomarker?

Answer: The Left Superior Medial Frontal to Right Middle Occipital edge was our #1 ranked biomarker (importance 1.000). It mediates visual-executive integration, directly reflecting the socio-visual gaze aversion and sensory gating atypicalities observed in autism.

Q14: How does DropEdge regularization prevent overfitting in GNNs?

Answer: DropEdge randomly removes 20% of graph edges during each training epoch (Rong et al., 2020), acting as message-passing dropout. This prevents the GAT from over-smoothing node representations across dense cliques and memorizing individual graph quirks.

Q15: What loss function and scheduler did you use?

Answer: We used Weighted Cross-Entropy Loss (weighted inversely by class frequencies) with Label Smoothing (0.05) to prevent overconfident misclassifications, optimized via CosineAnnealingWarmRestarts with AdamW.

Q16: Why did deep ResGAT (NB09) perform worse than Improved GAT (NB07)?

Answer: ResGAT introduced input projection layers and residual skip connections totaling 164,000+ parameters. On a medical dataset of 871 subjects, this caused over-parameterization and overfitting to training noise.

Q17: Can this model be deployed in a clinical hospital setting today?

Answer: It provides an automated clinician-support tool, not an autonomous replacement for psychiatrists. By presenting both a diagnostic probability and a visual biomarker subgraph, clinicians can inspect which brain circuits are atypical for a specific patient. Feature extraction and inference take <1.5 seconds per subject on standard CPU hardware.

Q18: What repetition time (TR) variations exist in ABIDE?

Answer: TR ranges from 1.5 seconds (e.g. Stanford) to 3.0 seconds (e.g. Leuven). This variation alters the temporal frequency resolution of BOLD time-series, which is why ComBat harmonization is vital.

Q19: How did you handle subjects with different time-series lengths?

Answer: Time-series length T varies from 78 to 320 timepoints across ABIDE sites. Because Pearson correlation and Ledoit-Wolf covariance normalize over temporal length T , the resulting connectivity representations are standardized 116×116 matrices.

Q20: What is the Fréchet mean in Riemannian geometry?

Answer: In Euclidean space, the mean is the arithmetic average $(1/N) \sum X_i$. On a curved Riemannian manifold, the Fréchet mean is the matrix that minimizes the sum of squared geodesic (curved) distances to all subjects' covariance matrices.

Q21: Why is Ledoit-Wolf shrinkage better than sample covariance?

Answer: When the number of brain regions ($P = 116$) is comparable to the number of temporal frames ($T \approx 150$ – 200), sample covariance matrices become ill-conditioned and noisy. Ledoit-Wolf asymptotically calculates the optimal shrinkage intensity towards a diagonal target, guaranteeing well-conditioned, invertible matrices.

Q22: What is the role of the Default Mode Network (DMN) in your findings?

Answer: Our #3 ranked biomarker (Anterior Cingulate to Inferior Temporal) is a core hub of the DMN and Salience Network. DMN dysregulation is widely associated with atypical self-referential cognition and theory-of-mind deficits in ASD.

Q23: How does your 68.88% full-dataset accuracy compare with published literature?

Answer: In rigorous, leak-free 10-fold cross-validation across all 17 multi-site ABIDE centers, 68%–70% is the established state-of-the-art benchmark (Dadi et al., NeuroImage 2020). Studies reporting >85% on all 871 subjects universally suffer from pre-split data leakage.

Q24: What are the main limitations of this study?

Answer: (1) The AAL-116 atlas is anatomically defined rather than functionally parcellated (like Schaefer or Glasser). (2) ABIDE I is predominantly male (85% male), reflecting real-world diagnostic biases. Future work will test female-specific connectome harmonizations.

Q25: What is the single most important takeaway of your research?

Answer: Multi-site fMRI classification requires a dual approach: Riemannian Geometric Harmonization to eliminate scanner noise and achieve high accuracy (68.88%–70.35%), combined with Explainable Graph Attention Networks to extract biologically validated, clinician-interpretable neural biomarkers (86.7% inter-lobar dysconnectivity).

Q26: How does BOLD signal power relate to autism neuropathology?

Answer: Node feature importance analysis revealed that BOLD signal power (energy) accounts for 48.6% of the GAT feature attribution. Atypical low-frequency signal energy reflects localized metabolic alterations and dysregulated microvascular neuro-vascular coupling in autistic cortical networks.

Q27: Why did you use Youden's J statistic for decision threshold calibration?

Answer: In medical screening, default 0.50 probability cutoffs are sub-optimal under noisy class distributions. Youden's J index ($J = \text{Sensitivity} + \text{Specificity} - 1$) objectively finds the optimal operating point on the ROC curve that maximizes both sensitivity and specificity simultaneously.

Q28: What are the 24 parameters in Friston's head motion model?

Answer: Friston's 24-parameter model includes the 6 rigid-body realignment parameters (3 translations, 3 rotations), their temporal derivatives (6 parameters), and the square of both the realignment parameters and their derivatives (12 parameters), effectively removing non-linear head motion artifacts.

Q29: How does your model handle negative functional correlations?

Answer: In the Riemannian tangent space formulation, all continuous covariance values (both positive and negative) are fully preserved. In the GAT graph construction, we focus on positive functional co-activations ($\tau = 0.20$ to 0.30) to preserve biologically coherent excitatory signaling pathways.

Q30: What is the future roadmap for this project?

Answer: Future milestones include: (1) Integrating dynamic time-varying functional connectivity, (2) Multi-atlas parcellation fusion (AAL + Harvard-Oxford + Schaefer-400), and (3) Prospective validation on ABIDE II and longitudinal developmental cohorts.

Part 4: Foundational Scientific References (30 Full Publications)

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